

Task 1

Problem: **Company Employee Access System**

Build a small employee access model for a company, applying encapsulation and inheritance.

1. Create class **Employee** with private fields:
 - **employeeId (int)** — auto-generated using a private static int counter that increments whenever a new Employee is created. Do not provide a setter for employeeId.
 - **name (String)**
 - **dailyAccessHours (int)** — default value 8.
 - Provide a **constructor that takes only name**. The employeeId must be generated automatically.
 - Provide **getters for all fields**.
 - Add a method **describe()** that prints the employee's ID, name, and allowed daily access hours.
 - If the provided name is blank or empty, the constructor should:
 - Print an error message.
 - Set the employee's name to "Unknown Employee".
2. Create subclass **Manager** extends Employee with an additional private field:
 - **teamSize (int)**
 - Its constructor takes **name** and **teamSize**.
Override **getDailyAccessHours()** so that a manager receives:
8 + (teamSize ~/ 5) hours
For example, a manager with a team size of 12 receives $8 + 2 = 10$ hours.
Override **describe()** to also display the manager's team size.
3. Create subclass **Intern** extends Employee with an additional private field:
 - **month (int)**
 - Its constructor takes **name** and **month**.
Override getDailyAccessHours():
 - Months 1–3 → 4 hours
 - Months 4–6 → 6 hours
4. In **Main**:
 - Create one plain Employee.
 - Create one Manager.
 - Create one Intern.
 - Call describe() for each object.
 - Print getDailyAccessHours() for each object.
 - Create another Employee with a blank name to demonstrate constructor validation.